9.4: During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis

Our main objective in this chapter is to learn how cells harvest the energy of glucose and other nutrients in food to make ATP. But the metabolic components of respiration we have examined so far, glycolysis and the citric acid cycle, produce only 4 ATP molecules per glucose molecule, all by substrate-level phosphorylation: 2 net ATP from glycolysis and 2 ATP from the citric acid cycle. At this point, molecules of NADH (and FADH2) account for most of the energy extracted from each glucose molecule. These electron escorts link glycolysis and the citric acid cycle to the machinery of oxidative phosphorylation, which uses energy released by the electron transport chain to power ATP synthesis. In this section, you will learn first how the electron transport chain works and then how electron flow down the chain is coupled to ATP synthesis.

The Pathway of Electron Transport

The electron transport chain is a collection of molecules embedded in the inner membrane of the mitochondrion in eukaryotic cells. (In prokaryotes, these molecules reside in the plasma membrane.) The folding of the inner membrane to form cristae increases its surface area, providing space for thousands of copies of each component of the electron transport chain in a mitochondrion. Structure fits function: The infolded membrane with its concentration of electron carrier molecules is well-suited for the series of sequential redox reactions that take place along the electron transport chain. Most components of the chain are proteins, which exist in multiprotein complexes numbered I through IV. Tightly bound to these proteins are *prosthetic groups*, nonprotein components such as cofactors and coenzymes essential for the catalytic functions of certain enzymes.

Figure 9.12 shows the sequence of electron carriers in the electron transport chain and the drop in free energy as electrons travel down the chain. During this electron transport, electron carriers alternate between reduced and oxidized states as they accept and then donate electrons. Each component of the chain becomes reduced when it accepts electrons from its “uphill” neighbor, which is a better electron donor. It then returns to its oxidized form as it passes electrons to its “downhill” neighbor, which is a better electron acceptor.

Figure 9.12

Free-energy change during electron transport.

Graphical representation of free energy change during electron transport. Y-axis is free energy (G) relative to O 2 (k cal per mol). Prosthetic groups in electron transport chain show a downhill slope toward O 2.

Overview: Steps in cellular respiration: glycolysis, pyruvate oxidation, citric acid cycle, and oxidative phosphorylation (produces A T P). All steps except glycolysis take place in the mitochondrion. 2 electrons from N A D H enter the electron transport chain, oxidizing N A D H to N A D plus. The electrons are passed downhill through three protein complexes, with several prosthetic groups bound in each. Flow of electrons from N A D H (least electronegative): Complex I: F M N to F E S. Between Complex 1 and 3 electrons pass to Q. In Complex 3 electrons flow from Cyt b to F E S to Cyt C sub 1. Between Complex 3 and 4 electrons pass to Cyt c. In Complex 4 electrons pass to Cyt a and then Cyt a sub 3. 2 electrons then join one-half O2 and 2 H plus (most electronegative) to form H 2 0. Complexes I-IV each consist of multiple proteins with electron carriers. 2 electrons from F A D H 2 can also join the chain from Complex 2, passing through F E S and then joining Q between Complexes 1 and 3, from which point they will follow the same path as electrons from N A D H.

Electron transport chain: Electrons (from NADH or F A D H 2) move from a less electronegative electron carrier (one with a lower affinity for electrons) to a more electronegative electron carrier down the chain, releasing free energy. The last electron carrier (Cyt a3) passes its electrons to oxygen, which is very electronegative.

The overall energy drop (ΔG) for electrons traveling from NADH to oxygen is 53 kcal/mol, but this “fall” is broken up into a series of smaller steps by the electron transport chain. (An oxygen atom is represented here as 12O2 to show that O2 is reduced, not individual oxygen atoms.) To view these proteins in their cellular context, see Figure 6.32b.

VISUAL SKILLS *Compare the position of the electrons in NADH (see*Figure 9.3*)* at the top of the chain with that in*H2O*, at the bottom. Describe why the electrons in*H2O*have less potential energy, using the term electronegativity.*

Figure 9.12 Walkthrough

Free-energy change during electron transport

Watch an explanation of this figure in the Figure 9.12 Walkthrough: Free-energy change during electron transport.

image

image

**

Now let's take a closer look at the electrons as they drop in energy level, passing through the components of the electron transport chain in Figure 9.12. We'll first look at the passage of electrons through complex I in some detail as an illustration of the general principles involved in electron transport. Electrons acquired from glucose by NAD+ during glycolysis and the citric acid cycle are transferred from NADH to the first molecule of the electron transport chain in complex I. This molecule is a flavoprotein, so named because it has a prosthetic group called flavin mononucleotide (FMN). In the next redox reaction, the flavoprotein returns to its oxidized form as it passes electrons to an iron-sulfur protein (Fe·S in complex I), one of a family of proteins with both iron and sulfur tightly bound. The iron-sulfur protein then passes the electrons to a compound called ubiquinone (Q in Figure 9.12). This electron carrier is a small hydrophobic molecule, the only member of the electron transport chain that is not a protein. Ubiquinone is individually mobile within the membrane rather than residing in a particular complex. (Another name for ubiquinone is coenzyme Q, or CoQ; you may have seen it sold as a nutritional supplement.)

Most of the remaining electron carriers between ubiquinone and oxygen are proteins called **cytochromes**. Their prosthetic group, called a heme group, has an iron atom that accepts and donates electrons. (The heme group in a cytochrome is similar to the heme group in hemoglobin, the protein of red blood cells, except that the iron in hemoglobin carries oxygen, not electrons.) The electron transport chain has several types of cytochromes, each named "cyt" with a letter and number to distinguish it as a different protein with a slightly different electron-carrying heme group. The last cytochrome of the chain, cyt a3, passes its electrons to oxygen (in O2), which is *very* electronegative. Each O also picks up a pair of hydrogen ions (protons) from the aqueous solution, neutralizing the −2 charge of the added electrons and forming water.

Another source of electrons for the electron transport chain is FADH2, the other reduced product of the citric acid cycle. Notice in Figure 9.12 that FADH2 adds its electrons from within complex II, at a lower energy level than NADH does. Consequently, although NADH and FADH2 each donate an equivalent number of electrons (2) for oxygen reduction, the electron transport chain provides about one-third less energy for ATP synthesis when the electron donor is FADH2 rather than NADH. We'll see why in the next section.

The electron transport chain makes no ATP directly. Instead, it eases the fall of electrons from food to oxygen, breaking a large free-energy drop into a series of smaller steps that release energy in manageable amounts, step by step. How does the mitochondrion (or the plasma membrane in prokaryotes) couple this electron transport and energy release to ATP synthesis? The answer is a mechanism called chemiosmosis.

Chemiosmosis: The Energy-Coupling Mechanism

Populating the inner membrane of the mitochondrion or the prokaryotic plasma membrane are many copies of a protein complex called **ATP synthase**, the enzyme that makes ATP from ADP and inorganic phosphate

(Figure 9.13). ATP synthase works like an ion pump running in reverse. Ion pumps usually use ATP as an energy source to transport ions against their gradients. Enzymes can catalyze a reaction in either direction, depending on the ΔG for the reaction, which is affected by the local concentrations of reactants and products (see Concepts 8.2 and 8.3). Under the conditions of cellular respiration, rather than hydrolyzing ATP to pump protons against their concentration gradient, ATP synthase uses the energy of an existing ion gradient to power ATP synthesis. The power source for ATP synthase is a difference in the concentration of H+ (a pH difference) on opposite sides of the inner mitochondrial membrane. This process, in which energy stored in the form of a hydrogen ion gradient across a membrane is used to drive cellular work such as the synthesis of ATP, is called **chemiosmosis** (from the Greek *osmos*, push). The word *osmosis* was previously used in discussing water transport, but here it refers to the flow of H+ across a membrane.

Figure 9.13

ATP synthase, a molecular mill.

A T P synthase in mitochondrial matrix, attached to rotor imbedded in the membrane through the internal rod and the catalytic rod; stator adjacent to the rotor open to the intermembrane space.

H plus ions flowing down their gradient enter a channel in a stator, which is anchored in the membrane. H plus ions are concentrated in the intermembrane space until they enter the stator. There is a clear concentration gradient of H plus ions across the mitochondrial membrane. H plus ions enter binding sites within a rotor, changing the shape of each subunit so that the rotor spins within the membrane. Each H plus ion makes one complete turn before leaving the rotor and passing through a second channel in the stator into the mitochondrial matrix. The spinning of the rotor causes an internal rod to spin as well. This rod extends like a stalk into the knob below it, which is held stationary by part of the stator. Turning the rod activates catalytic sites in the knob that produce A T P from A D P and P sub i.

Multiple ATP synthases reside in eukaryotic mitochondrial and chloroplast membranes and in prokaryotic plasma membranes. (See Figure 6.32b and c.)

BioFlix® Animation

ATP Synthase

image

image

**

Animation

Rotating ATP Synthase

image

image

**

From studying the structure of ATP synthase, scientists have learned how the flow of H+ through this enzyme powers ATP generation. ATP synthase is a multisubunit complex with four main parts, each made up of multiple polypeptides (see Figure 9.13). Protons move one by one into binding sites on the rotor, causing it to spin in a way that catalyzes ATP production from ADP and Ⓟi. The flow of protons thus behaves somewhat like a rushing stream that turns a waterwheel. ATP synthase is the smallest molecular rotary motor known in nature.

How does the inner mitochondrial membrane (or the prokaryotic plasma membrane) generate and maintain the H+ gradient that drives ATP synthesis by the ATP synthase protein complex? Establishing the H+ gradient is a major function of the electron transport chain, which is shown in its mitochondrial location in **Figure 9.14**. The chain is an energy converter that uses the exergonic flow of electrons from NADH and FADH2 to pump H+ across the membrane, from the mitochondrial matrix into the intermembrane space. The H+ has a tendency to move back across the membrane, diffusing down its gradient. And the ATP synthases are the only sites that provide a route through the membrane for H+. As we described previously, the passage of H+ through ATP synthase uses the exergonic flow of H+ to drive the phosphorylation of ADP. Thus, the energy stored in an H+ gradient across a membrane couples the redox reactions of the electron transport chain to ATP synthesis.

Figure 9.14

Chemiosmosis couples the electron transport chain to ATP synthesis.

In the electron transport chain, electron transport through a series of protons embedded in the inner mitochondrial membrane causes protons to be pumped, creating an H plus gradient across the membrane.

Inner mitochondrial membrane separates mitochondrial matrix from intermembrane space. N A D H carries electrons from food to the electron transport chain, a series of protein complexes of electron carriers embedded in the inner mitochondrial membrane. The electrons are passed through Complex 1, and a proton is pumped into the intermembrane space. The electrons then pass downhill to Q, then to Complex 3, where a proton is pumped into the intermembrane space. The electrons pass through Cyt c and then to Complex 4,

where a proton is pumped into the intermembrane space, and at the bottom of this last complex, electrons join 2 H plus and one-half O 2 to form H 2 O. Electrons from F A D H 2 can also join the electron transport chain at Q, from which point they will follow the same path as electrons from N A D H.

Chemiosmosis: A T P synthase is powered by the flow of H plus back across the inner mitochondrial membrane, forming A T P. Protons pumped into the intermembrane space during the electron transport chain fall back down their concentration gradient through the A T P synthase (embedded in the inner mitochondrial membrane), causing inorganic phosphate to be added to A D P, forming A T P.

STEP 1 NADH and FADH2 shuttle high-energy electrons extracted from food during glycolysis and the citric acid cycle into an electron transport chain built into the inner mitochondrial membrane. (See Figure 6.32b.) The gold arrows trace the transport of electrons, which are finally passed to a terminal acceptor (O2, in the case of aerobic respiration) at the “downhill” end of the chain, forming water. Most of the electron carriers of the chain are grouped into four complexes (I–IV). Two mobile carriers, ubiquinone (Q) and cytochrome c (Cyt c), move rapidly, ferrying electrons between the large complexes. As the complexes shuttle electrons, they pump protons from the mitochondrial matrix into the intermembrane space. FADH2 deposits its electrons via complex II and so results in fewer protons being pumped into the intermembrane space than occurs with NADH. Chemical energy originally harvested from food is transformed into a proton-motive force, a gradient of H+ across the membrane. **STEP 2** During chemiosmosis, the protons flow back down their gradient via ATP synthase, which is built into the membrane nearby. The ATP synthase harnesses the proton-motive force to phosphorylate ADP, forming ATP. Together, electron transport and chemiosmosis make up oxidative phosphorylation.

WHAT IF? *If complex IV were nonfunctional, could chemiosmosis produce any**ATP**, and if so, how would the rate of synthesis differ?*

*For suggested answer, see*Appendix A.**

Animation

Electron Transport

image

image

**

BioFlix® Animation

Electron Transport

image

image

**

At this point, you may be wondering how the electron transport chain pumps hydrogen ions into the intermembrane space. Researchers have found that certain members of the electron transport chain accept and release protons H+ along with electrons. (The aqueous solutions inside and surrounding the cell are a ready source of H+.) At certain steps along the chain, electron transfers cause H+ to be taken up and released into the surrounding solution. In eukaryotic cells, the electron carriers are spatially arranged in the inner mitochondrial membrane in such a way that H+ is accepted from the mitochondrial matrix and deposited in the intermembrane space (see Figure 9.14). The H+ gradient that results is referred to as a **proton-motive force**, emphasizing the capacity of the gradient to perform work. The force drives H+ back across the membrane through the H+ channels provided by ATP synthases.

In general terms, *chemiosmosis is an energy-coupling mechanism that uses energy stored in the form of an*H+*gradient across a membrane to drive cellular work*. In mitochondria, the energy for gradient formation comes from exergonic redox reactions along the electron transport chain, and ATP synthesis is the work performed. But chemiosmosis also occurs elsewhere and in other variations. Chloroplasts use chemiosmosis to generate ATP during photosynthesis; in these organelles, light (rather than chemical energy) drives both electron flow down an electron transport chain and the resulting H+ gradient formation. Prokaryotes, as already mentioned, generate H+ gradients across their plasma membranes. They then tap the proton-motive force not only to make ATP inside the cell but also to rotate their flagella and to pump nutrients and waste products across the membrane. Because of its central importance to energy conversions in prokaryotes and eukaryotes, chemiosmosis has helped unify the study of bioenergetics. Peter Mitchell was awarded the Nobel Prize in 1978 for originally proposing the chemiosmotic model.

An Accounting of ATP Production by Cellular Respiration

In the last few sections, we have looked rather closely at the key processes of cellular respiration. Now let's take a step back and remind ourselves of its overall function: harvesting the energy of glucose for ATP synthesis.

During respiration, most energy flows in this sequence:
glucose→NADH→electron transport chain→protonmotive force→ATP. We can do some bookkeeping to calculate the ATP profit when cellular respiration oxidizes a molecule of glucose to six molecules of carbon dioxide. The three main departments of this metabolic enterprise are glycolysis, pyruvate oxidation and the citric acid cycle, and the electron transport chain, which drives oxidative phosphorylation. **Figure 9.15** gives a detailed accounting of the ATP yield for each glucose molecule that is oxidized. The tally adds the

4 ATP produced directly by substrate-level phosphorylation during glycolysis and the citric acid cycle to the many more molecules of ATP generated by oxidative phosphorylation. Each NADH that transfers a pair of electrons from glucose to the electron transport chain contributes enough to the proton-motive force to generate a maximum of about 3 ATP.

Figure 9.15

ATP yield per molecule of glucose at each stage of cellular respiration.

A T P yield per molecule of glucose at each stage of cellular respiration.

In the cytosol, glycolysis produces 2 A T P via substrate-level phosphorylation; 2 pyruvates from glucose which enter pyruvate oxidation (inside the mitochondrion); and 2 NADH which carry electrons to oxidative phosphorylation inside the mitochondrion. In pyruvate oxidation, 2 pyruvates (from glycolysis) are converted to Acetyl Co-A, which will go to the citric acid cycle. 2 N A D H also carry electrons to oxidative phosphorylation. In the citric acid cycle, 2 A T P are formed, ad 6 N A D H and 2 F A D H 2 carry electrons to oxidative phosphorylation. Oxidative phosphorylation produces about 26 or 28 A T P by oxidative phosphorylation, depending on which shuttles transports electrons from N A D H in the cytosol. The maximum per glucose formation of A T P is about 30 or 32 A T P.

VISUAL SKILLS *After reading the discussion in the text, explain exactly how the total of* 26 *or* 28 ATP *from oxidative phosphorylation was calculated (see the yellow bar).*

Animation

ATP Yield from Cellular Respiration

image

image

**

Why are the numbers in Figure 9.15 inexact? There are three reasons we cannot state an exact number of ATP molecules generated by the breakdown of one molecule of glucose. First, phosphorylation and the redox reactions are not directly coupled to each other, so the ratio of the number of NADH molecules to the number of ATP molecules is not a whole number. We know that 1 NADH results in 10 H+ being transported out across the inner mitochondrial membrane, but the exact number of H+ that must reenter the mitochondrial matrix via ATP synthase to generate 1 ATP has long been debated. Based on experimental data, however, most biochemists now agree that the most accurate number is 4 H+. Therefore, a single molecule of NADH generates enough proton-motive force for the synthesis of 2.5 ATP. The citric acid cycle also supplies electrons to the electron transport chain via FADH2, but since its electrons enter later in the

chain, each molecule of this electron carrier is responsible for transport of only enough H+ for the synthesis of 1.5 ATP. These numbers also take into account the slight energetic cost of moving the ATP formed in the mitochondrion out into the cytosol, where it will be used.

Second, the ATP yield varies slightly depending on the type of shuttle used to transport electrons from the cytosol into the mitochondrion. The mitochondrial inner membrane is not permeable to NADH, so NADH in the cytosol is segregated from the machinery of oxidative phosphorylation. The 2 electrons of NADH captured in glycolysis must be conveyed into the mitochondrion by one of several electron shuttle systems. Depending on the kind of shuttle in a particular cell type, the electrons are passed either to NAD+ or to FAD in the mitochondrial matrix (see Figures 9.14 and 9.15). If the electrons are passed to FAD, as in brain cells, only about 1.5 ATP can result from each NADH that was originally generated in the cytosol. If the electrons are passed to mitochondrial NAD+, as in liver cells and heart cells, the yield is about 2.5 ATP per NADH.

A third variable that reduces the yield of ATP is the use of the proton-motive force generated by the redox reactions of respiration to drive other kinds of work. For example, the proton-motive force powers the mitochondrion's uptake of pyruvate from the cytosol (see Figure 9.9). However, if *all* the proton-motive force generated by the electron transport chain were used to drive ATP synthesis, one glucose molecule could generate a maximum of 28 ATP produced by oxidative phosphorylation plus 4 ATP (net) from substrate-level phosphorylation to give a total yield of about 32 ATP (or only about 30 ATP if the less efficient shuttle were functioning).

Animation

Electron Transport Chain: Factors Affecting ATP Yield

image

**

We can now roughly estimate the efficiency of respiration—that is, the percentage of chemical energy in glucose that has been transferred to ATP. Recall that the complete oxidation of a mole of glucose releases 686 kcal of energy under standard conditions ($\Delta G = -686$ kcal/mol). Phosphorylation of ADP to form ATP stores at least 7.3 kcal per mole of ATP. Therefore, the efficiency of respiration is 7.3 kcal per mole of ATP times 32 moles of ATP per mole of glucose divided by 68 kcal per mole of glucose, which equals 0.34. Thus, about 34% of the potential chemical energy in glucose has been transferred to ATP; the actual percentage is bound to vary as ΔG varies under different cellular conditions. Cellular respiration is remarkably efficient in its energy conversion. By comparison, even the most efficient automobile converts only about 25% of the energy stored in gasoline to energy that moves the car.

The rest of the energy stored in glucose is lost as heat. We humans use some of this heat to maintain our relatively high body temperature (37°C), and we dissipate the rest through sweating and other cooling

mechanisms.

Surprisingly, perhaps, it may be beneficial under certain conditions to reduce the efficiency of cellular respiration. A remarkable adaptation is shown by hibernating mammals, which overwinter in a state of inactivity and lowered metabolism. Although their internal body temperature is lower than normal, it still must be kept significantly higher than the external air temperature. One type of tissue, called brown fat, is made up of cells packed full of mitochondria. The inner mitochondrial membrane contains a channel protein called the uncoupling protein that allows protons to flow back down their concentration gradient without generating ATP. Activation of these proteins in hibernating mammals results in ongoing oxidation of stored fuel (fats), generating heat without any ATP production. In the absence of such an adaptation, the buildup of ATP would eventually cause cellular respiration to be shut down by regulatory mechanisms that will be discussed later. Brown fat is also used for heat generation in humans. In the **Scientific Skills Exercise**, you can work with data in a related but different case where a decrease in metabolic efficiency in cells is used to generate heat.

Scientific Skills Exercise

Making a Bar Graph and Evaluating a Hypothesis

AP Science Practices 1, 2, 3, 4

Does Thyroid Hormone Level Affect O_2 Consumption in Cells? Some animals, such as mammals and birds, maintain a relatively constant body temperature, above that of their environment, by using heat produced as a by-product of metabolism. When the core temperature of these animals drops below an internal set point, their cells are triggered to reduce the efficiency of ATP production by the electron transport chains in mitochondria. At lower efficiency, extra fuel must be consumed to produce the same number of ATPs, generating additional heat. This response is moderated by the endocrine system, and researchers hypothesized that it might be triggered by thyroid hormone. In this exercise, you will use a bar graph to visualize data from an experiment that compared the metabolic rates (by measuring O_2 consumption) in mitochondria of cells from animals with different levels of thyroid hormone.

A mouse.

How the Experiment Was Done Liver cells were isolated from sibling rats that had low, normal, or elevated thyroid hormone levels. The oxygen consumption rate due to activity of the mitochondrial electron transport chains of each type of cell was measured under controlled conditions.

Data from the Experiment

A table lists thyroid hormone levels and oxygen consumption rate.

Data from M. E. Harper and M. D. Brand, The quantitative contributions of mitochondrial proton leak and ATP turnover reactions to the changed respiration rates of hepatocytes from rats of different thyroid status, *Journal of Biological Chemistry* 268:14850–14860 (1993).

Thyroid hormone level	Oxygen consumption rate
Low	4.3
Normal	4.8
Elevated	8.7

INTERPRET THE DATA

- To visualize any differences in O2 consumption between cell types, it will be useful to graph the data in a bar graph. First, set up the axes.
- What is the independent variable (intentionally varied by the researchers), which goes on the x-axis? List the categories along the x-axis; because they are discrete rather than continuous, you can list them in any order.
- What is the dependent variable (measured by the researchers), which goes on the y-axis?
- What units (abbreviated) should go on the y-axis? Label the y-axis, including the units specified in the data table. Determine the range of values of the data that will need to go on the y-axis. What is the largest value? Draw evenly spaced tick marks and label them, starting with 0 at the bottom.
- Graph the data for each sample. Match each x-value with its y-value and place a mark on the graph at that coordinate, then draw a bar from the x-axis up to the correct height for each sample. Why is a bar graph more appropriate than a scatter plot or line graph? (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Examine your graph and look for a pattern in the data.
- Which cell type had the highest rate of O2 consumption, and which had the lowest?
- Does this support the researchers' hypothesis? Explain.
- Based on what you know about mitochondrial electron transport and heat production, predict which rats had the highest, and which had the lowest, body temperature.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

BioFlix® Animation

Cellular Respiration

**